

1 Theory

Lensing formalism

Lectures on gravitational lensing (Narayan and Bartelmann)

galaxiesbook

Gravitational lensing is the phenomenon that describes how the path of light bends in the presence of a gravitational field. It is one of the most famous predictions from Einstein's theory of general relativity. Generally, people categorise lensing events into:

- Strong lensing: large masses with favourable geometries. We see two or more separate images of the source
- Weak lensing: large masses with less favourable geometry. we see sheared or sheared images
- Microlensing: involves small masses where we can't resolve the images from one another. We instead see a brightening of the source

Assuming that a lens has circular symmetry (not necessarily always true), from general relativity we find that the deflection angle α is given by

$$\alpha(\xi) = \frac{4GM}{c^2} \frac{1}{\xi}$$

Where $M(\xi)$ is the mass contained within radius ξ . In the small angle approximation,

$$D_{LS}\alpha = D_S(\theta - \beta)$$

$$\xi = D_L\theta$$

and in the perfectly symmetrical case where S lies directly behind L, so that $\beta = 0$, we have

$$\theta_E = \sqrt{\frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}}$$

In this situation the light from the source is smeared out into an Einstein ring with radius θ_E . If all lenses were point masses, we would get a ring for $\beta = 0$, otherwise we'd get 2 images. The magnification μ is the ratio between the solid angles of the image and the source

$$\mu = \frac{\theta}{\beta} \frac{d\theta}{d\beta}$$

and using the lens equation we find

$$\mu = \frac{u^2 + 2}{2u\sqrt{u^2 + 4}}$$

where $u = \beta/\theta_E$

Caustics

The lens equation is

$$\beta = \theta - \alpha(\theta)$$

It essentially connects two planes: θ - the apparent angular position of an image on the sky and the source plane (where the object actually is)

β - the true position of a source with no lensing

α - deflection caused by the gravitational lens

This connection (mapping) is non linear, so creates many things such as arcs, rings and multiple images. The **Jacobian** of lens mapping is

$$A = \frac{\partial \beta}{\partial \theta}$$

which we can derive from the lens equation if we really wanted to. We can find the magnification from

$$\mu = \frac{1}{\det A}$$

so the magnification becomes infinite when the determinant is 0. Curves in the image plane where $\det(a) = 0$ are called critical curves. If we map these critical curves through the lens equation into the source plane the resulting curves are called caustics. Near the caustics, the amplification is described by Airy (fold) or Pearcey (cusp) integrals

2 project outline?

The idea here is to solve the lens equation numerically. We could simulate an artificial source behind a galaxy, where images would arrive with different time delays

- 1. lens equation in 1D (along a line) is

$$\beta = \theta - \theta_E^2/\theta$$

and for a given β we can have two solutions

- 2. we solve the lens equation in 1D for θ using Newton-Raphson.

$$f(\theta) = \theta - \frac{\theta_E^2}{\theta} - \beta = 0$$

NR iterates as

$$\theta_{n+1} = \theta_n - \frac{f(\theta_n)}{f'(\theta_n)}$$

and finds our two image positions. (we could also make some nice example plots here using relatively simple python plotter??)

- 3.

$$t(\theta) = \frac{D}{c} \left(\frac{1}{2}(\theta - \beta)^2 - \psi(\theta) \right)$$

geometric plus gravitational (shapiro) delay. for point mass we take

$$\psi(\theta) = \theta_E \ln \theta$$

and we compute for each $\theta_{\pm}$

- 4. compute Δt between two images
- 5. model gaussian pulse every T